`saturation(T, comp)`

How the saturation pressure is found

PC-SAFT package — `pcsaft/equilibrium.py`

1 What it does

	symbol	units
Input	T, $\quad \boldsymbol{\theta} = (m,\ \sigma,\ \varepsilon/k)$	K; –, Å, K
Output	$P^{\mathrm{sat}}(T;\boldsymbol{\theta})$, $\rho_L(T;\boldsymbol{\theta})$, $\rho_V(T;\boldsymbol{\theta})$ or `None` if $T > T_c$	Pa; Å^{-3}, Å^{-3}

It solves, for the single unknown P,

$$\boxed{\ln\varphi\big(\eta_L(P,T;\boldsymbol{\theta}),\,T;\boldsymbol{\theta}\big) \;=\; \ln\varphi\big(\eta_V(P,T;\boldsymbol{\theta}),\,T;\boldsymbol{\theta}\big)} \tag{1}$$

No experimental value enters: every bracket comes from the equation of state.

2 Building blocks

Everything depends on the packing fraction η, the temperature T and the parameters $\boldsymbol{\theta}$.

$$d(T;\boldsymbol{\theta}) = \sigma\left[1 - 0.12\exp\left(-\frac{3\,\varepsilon/k}{T}\right)\right] \qquad [\text{Å}] \tag{2}$$

$$\rho(\eta,T;\boldsymbol{\theta}) = \frac{6\,\eta}{\pi\, m\, d(T;\boldsymbol{\theta})^3} \qquad [\text{Å}^{-3}] \tag{3}$$

$$Z(\eta,T;\boldsymbol{\theta}) = 1 + Z_{\mathrm{hc}}(\eta;m) + Z_{\mathrm{disp}}(\eta,T;\boldsymbol{\theta}) \qquad [-] \tag{4}$$

$$P_{\mathrm{EoS}}(\eta,T;\boldsymbol{\theta}) = 10^{30}\,\rho(\eta,T;\boldsymbol{\theta})\,k_B\,T\,Z(\eta,T;\boldsymbol{\theta}) \qquad [\text{Pa}] \tag{5}$$

$$\ln\varphi(\eta,T;\boldsymbol{\theta}) = \tilde{a}^{\mathrm{res}}(\eta,T;\boldsymbol{\theta}) + Z(\eta,T;\boldsymbol{\theta}) - 1 - \ln Z(\eta,T;\boldsymbol{\theta}) \qquad [-] \tag{6}$$

The factor 10^{30} converts Å^{-3} to m^{-3}, so that $\mathrm{m}^{-3}\cdot\mathrm{J\,K^{-1}}\cdot\mathrm{K} = \mathrm{Pa}$.

3 Algorithm

Step 1 — Spinodal `spinodal()`

On a log grid η_k, $k = 1,\dots,600$, from 10^{-10} to $\eta_{\max} = \pi/(3\sqrt{2}) \approx 0.7405$:

$$P_k = P_{\mathrm{EoS}}(\eta_k,T;\boldsymbol{\theta}), \qquad \Delta P_k = P_{k+1} - P_k \tag{7}$$

A turning point is any k with $\Delta P_{k-1}\,\Delta P_k < 0$. The first is the maximum, the second the minimum:

$$\eta_{\mathrm{sp},V}(T;\boldsymbol{\theta}),\quad P_{\max}(T;\boldsymbol{\theta}) \qquad\qquad \eta_{\mathrm{sp},L}(T;\boldsymbol{\theta}),\quad P_{\min}(T;\boldsymbol{\theta}) \tag{8}$$

Fewer than two turning points $\Rightarrow T > T_c \Rightarrow$ return `None`.

Step 2 — Densities at a trial pressure `_both_roots()`

For a given P, define the inner residual

$$f(\eta; P, T, \boldsymbol{\theta}) = P_{\mathrm{EoS}}(\eta, T; \boldsymbol{\theta}) - P \tag{9}$$

and solve it twice with `brentq`, each time inside a bracket supplied by Step 1:

$$\eta_V(P, T; \boldsymbol{\theta}): \quad f = 0 \ \text{ on } \ \big[\eta_{\mathrm{floor}}(P, T; \boldsymbol{\theta}),\ \eta_{\mathrm{sp},V}\big] \tag{10}$$

$$\eta_L(P, T; \boldsymbol{\theta}): \quad f = 0 \ \text{ on } \ \big[\eta_{\mathrm{sp},L},\ \eta_{\max}\big] \tag{11}$$

The lower edge is one tenth of the ideal-gas packing fraction, which always lies below the vapour root because $Z_V < 1$:

$$\eta_{\mathrm{floor}}(P, T; \boldsymbol{\theta}) = 0.1 \cdot \frac{\pi}{6}\, m\, d(T; \boldsymbol{\theta})^3\, \frac{P}{10^{30}\, k_B T} \tag{12}$$

These brackets are valid only for $P_{\min} < P < P_{\max}$.

Step 3 — Outer residual `_delta_ln_phi()`

$$g(P; T, \boldsymbol{\theta}) = \ln \varphi\big(\eta_L(P, T; \boldsymbol{\theta}),\, T; \boldsymbol{\theta}\big) - \ln \varphi\big(\eta_V(P, T; \boldsymbol{\theta}),\, T; \boldsymbol{\theta}\big) \tag{13}$$

$$g > 0: \ P < P^{\mathrm{sat}} \text{ (vapour stable)}, \qquad g = 0: \ P = P^{\mathrm{sat}}, \qquad g < 0: \ P > P^{\mathrm{sat}} \text{ (liquid stable)} \tag{14}$$

Each evaluation of g runs the two inner solves of Step 2.

Step 4 — Pressure window

$$P_{\mathrm{hi}}(T; \boldsymbol{\theta}) = \big(1 - 10^{-6}\big)\, P_{\max}(T; \boldsymbol{\theta}), \qquad P_{\mathrm{fl}}(T; \boldsymbol{\theta}) = \max\big(P_{\min}(T; \boldsymbol{\theta}),\, 0\big) \tag{15}$$

P_{hi} stays just below $P_{\max}$: at $P_{\max}$ the vapour root merges with $\eta_{\mathrm{sp},V}$ and its bracket has no sign change. P_{fl} is not used directly as the lower bracket: near $P = 0$ the liquid root sits on the near-vertical branch, $Z_L \to 0$ loses its significant figures, and $\ln Z_L$ can return NaN. Instead, walk up from it:

$$P_j = P_{\mathrm{fl}} + (P_{\mathrm{hi}} - P_{\mathrm{fl}})\, s_j, \qquad s_j \in \{10^{-12},\ \ldots,\ 0.5\} \ \text{ (60 values, log-spaced)} \tag{16}$$

$$P_{\mathrm{lo}} = \text{first } P_j \text{ with } g(P_j; T, \boldsymbol{\theta}) \text{ finite and } > 0 \tag{17}$$

If none qualifies, return `None`.

Step 5 — Saturation pressure

$$P^{\mathrm{sat}}(T; \boldsymbol{\theta}): \quad g\big(P; T, \boldsymbol{\theta}\big) = 0 \ \text{ on } \ \big[P_{\mathrm{lo}},\ P_{\mathrm{hi}}\big] \tag{18}$$

solved with `brentq`. The bracket is guaranteed: $g(P_{\mathrm{lo}}) > 0$ by construction and $g(P_{\mathrm{hi}}) < 0$.

Step 6 — Coexisting densities

$$\rho_L(T; \boldsymbol{\theta}) = \rho\Big(\eta_L\big(P^{\mathrm{sat}}, T; \boldsymbol{\theta}\big),\, T; \boldsymbol{\theta}\Big), \qquad \rho_V(T; \boldsymbol{\theta}) = \rho\Big(\eta_V\big(P^{\mathrm{sat}}, T; \boldsymbol{\theta}\big),\, T; \boldsymbol{\theta}\Big) \tag{19}$$

Returned as `(P_sat, rho_l, rho_v)`: liquid before vapour.

4 Flow chart

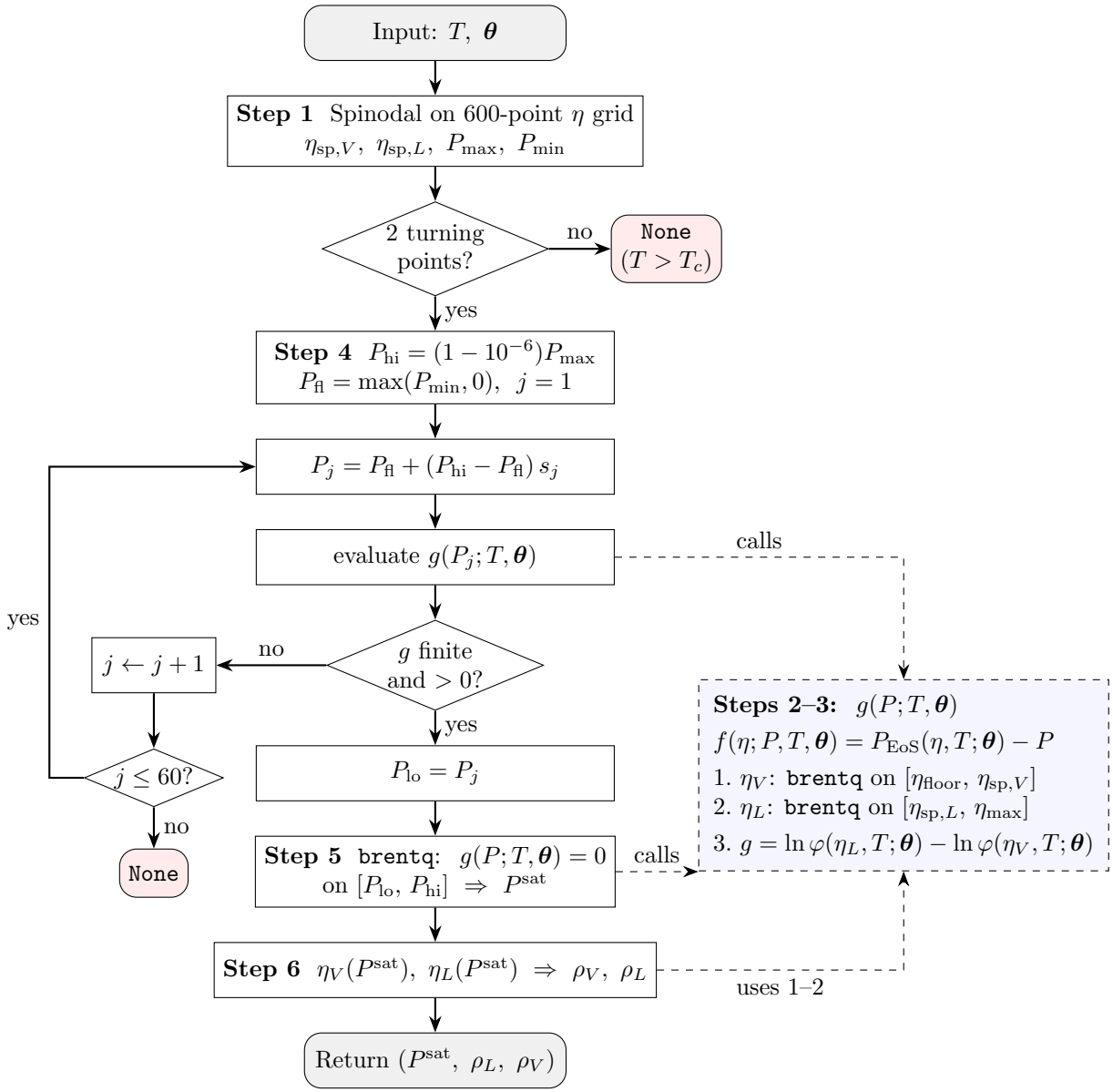


Figure 1: Two nested solvers. The outer `brentq` moves P; every trial P triggers two inner `brentq` calls on η. All brackets come from the spinodal.

5 Code map

function	computes	step
`spinodal(T, comp)`	$\eta_{\mathrm{sp},V},\ \eta_{\mathrm{sp},L},\ P_{\max},\ P_{\min}$	1
`_both_roots(T, P, comp, eta_v, eta_l)`	$\rho\big(\eta_V(P, T; \boldsymbol{\theta})\big),\ \rho\big(\eta_L(P, T; \boldsymbol{\theta})\big)$	2
`_delta_ln_phi(P, T, comp, eta_v, eta_l)`	$g(P; T, \boldsymbol{\theta})$	3
`saturation(T, comp)`	$P^{\mathrm{sat}},\ \rho_L,\ \rho_V$	4–6

6 Check: ethane at 250 K

$\boldsymbol{\theta} = (1.6069,\ 3.5206\ \text{Å},\ 191.42\ \text{K})$, $d(250\ \text{K}; \boldsymbol{\theta}) = 3.4781\ \text{Å}$.

Step 1		Steps 4–6	
$\eta_{\text{sp},V}$	0.05403	P_{lo}	2.39×10^{-6} Pa $(s_1 = 10^{-12})$
$\eta_{\text{sp},L}$	0.25596	P^{sat}	13.0388 bar
$P_{\max}$	23.913 bar	η_L, η_V	0.32048, 0.016481
$P_{\min}$	−92.071 bar	ρ_L, ρ_V	452.03, 23.246 kg m^{-3}

P / bar	0.01	1	10	13.0388	16	20
$g(P; 250\ \text{K}, \boldsymbol{\theta})$	+6.958	+2.368	+0.213	≈ 0	−0.151	−0.298

7 Limits

- The spinodal is a grid point, not the exact turning point, so $P_{\max}$ comes out slightly low and $P_{\min}$ slightly high. Close to T_c, where the loop is narrow, this is the first place to look if `saturation` fails.
- Each outer iteration costs two inner solves. The cost of one call is roughly one spinodal grid (600 evaluations of P_{EoS}) plus a few dozen `brentq` iterations.